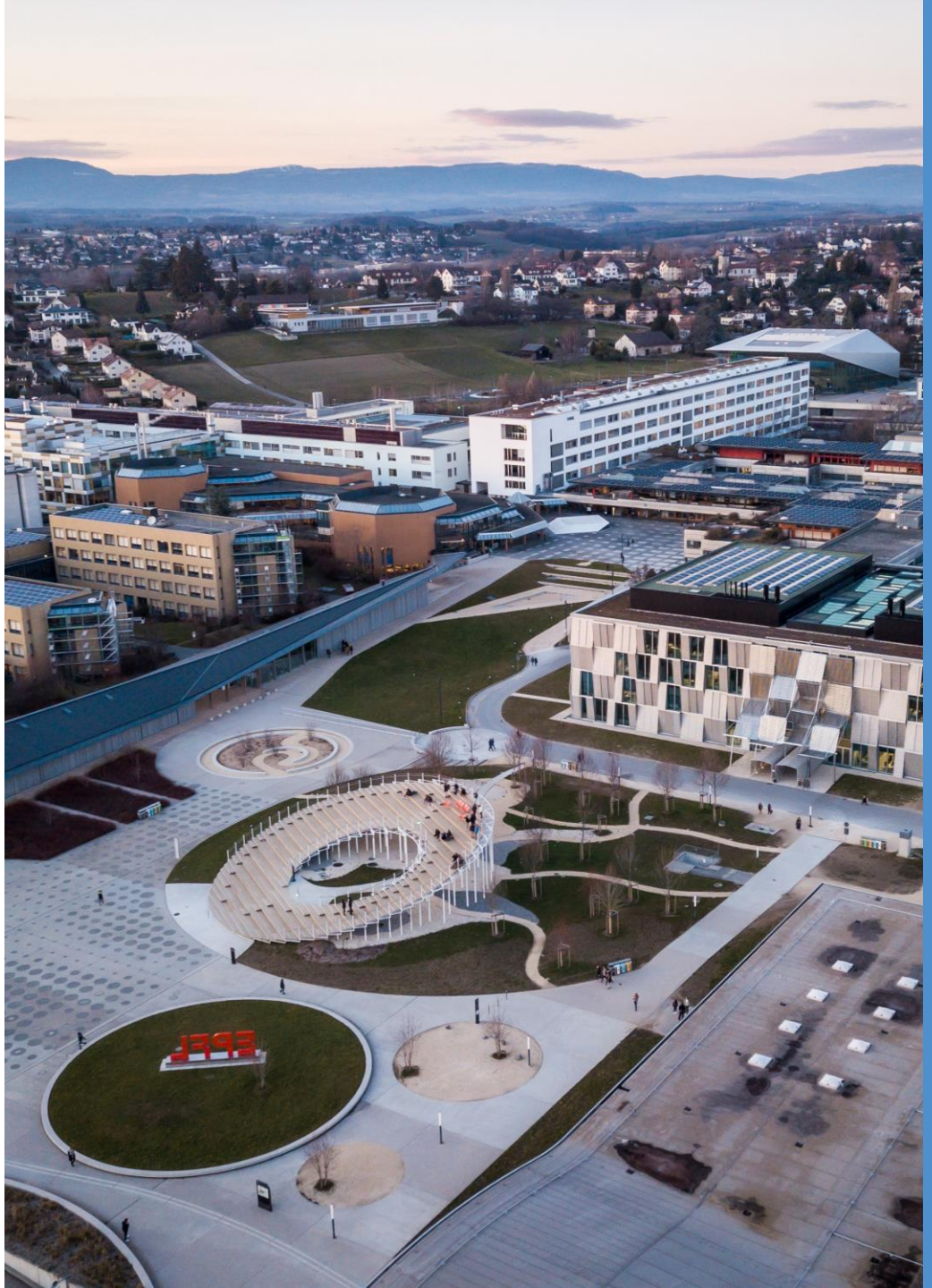


Methanol as a CO_2 Utilization Route

Meire Ribeiro Domingos
Daniel Florez-Orrego
Oktay Boztas



September 2026



OUTLINE

- Methanol as platform chemical and energy carrier
- Feedstock, carbon origin and end-of-life
- Methanol synthesis and heat management
- Industrial projects
- Production costs and challenges
- Industrial case studies

Why methanol?

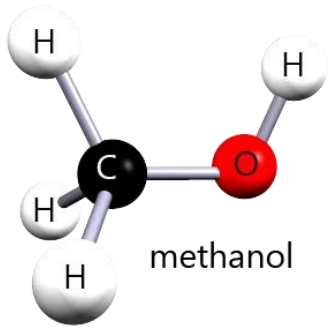
✓ **The simplest alcohol**

Freezing/Boiling

-97 °C, 65 °C

Liquid at 25 °C / 1 bar

12.6 wt.% H (19.9 MJ/kg)

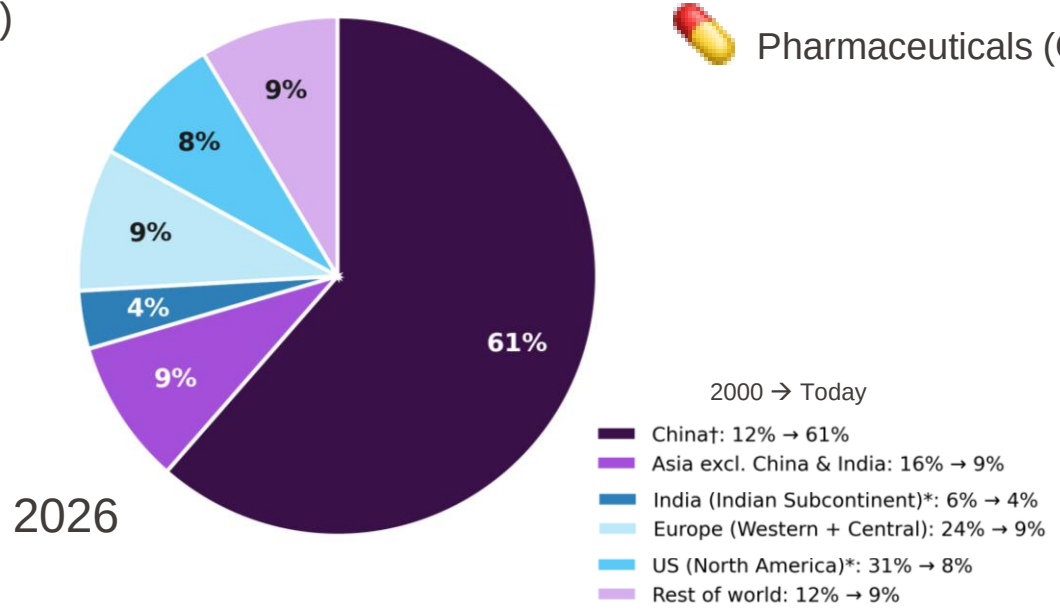


➢ **Global consumption** > 120 Mt (41B USD/y)

4% annual growth

90+ plants and **developed logistics**

One of the fifth largest bulk chemicals



2026

❑ **Strategic for the energy transition**

A platform chemical (C) and an energy carrier (F):

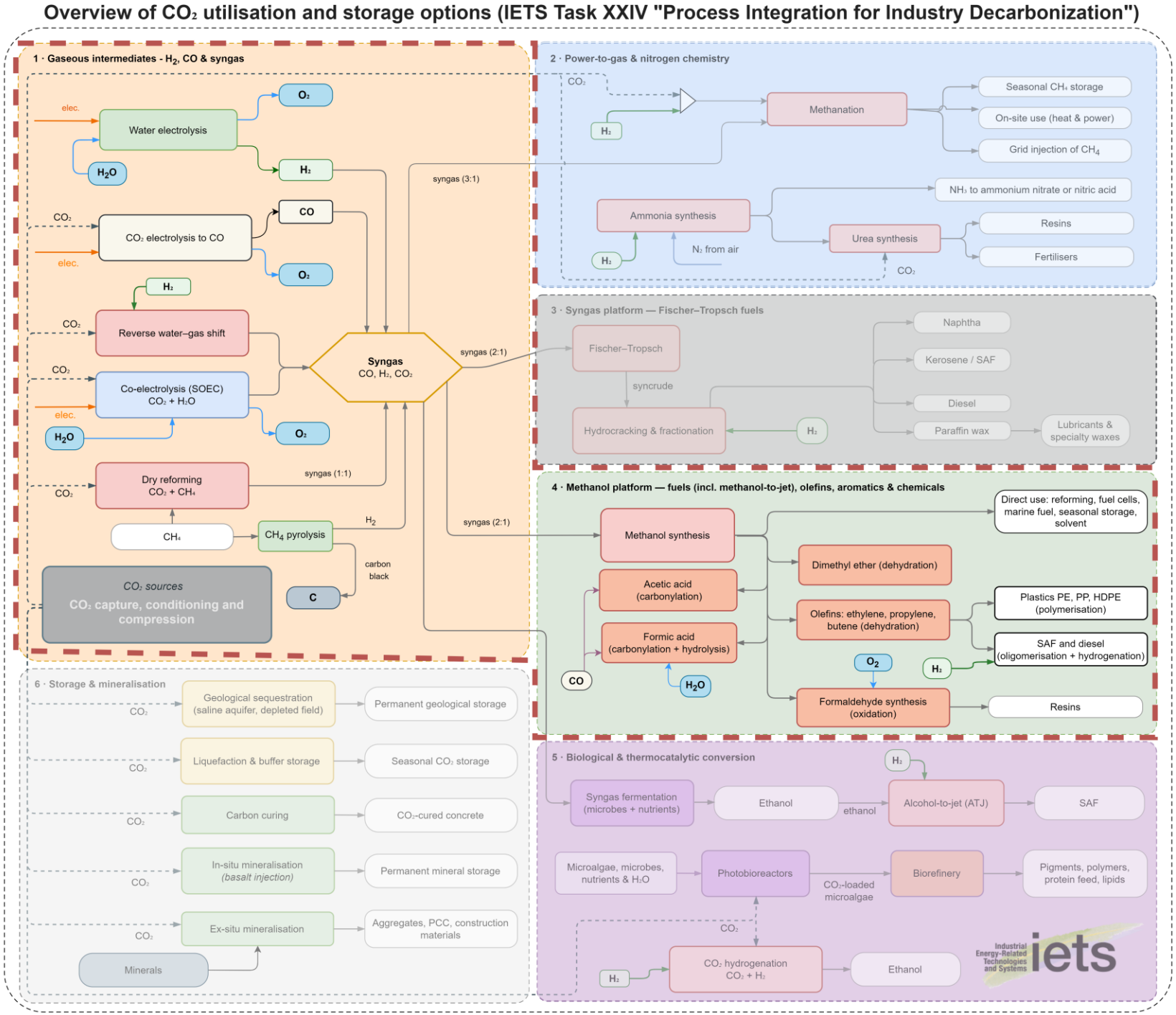
- Shipping and blended fuels (F)
- Chemicals feedstock (C)
- Solvents for paints and coatings (C)
- Antifreezers and refrigerants (F)
- Power-to-x and energy storage (F)
- Fuel-cell applications (F)
- Biodiesel production (C)
- Pharmaceuticals (C)

Sources:

Alvarado (2016), IHS Chemical Bulletin, Issue 3
IRENA (2021), Innovation Outlook: Renewable Methanol

CO_2 utilization and storage options

1 Gaseous intermediates: H_2, CO, syngas
2 Power-to-gas and nitrogen chemistry
3 Syngas to Fischer–Tropsch fuels
4 Methanol platform
5 Biological and thermocatalytic conversion
6 Storage and mineralisation



Industrial Energy-Related Technologies and Systems iets

Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Federal Office of Energy SFOE

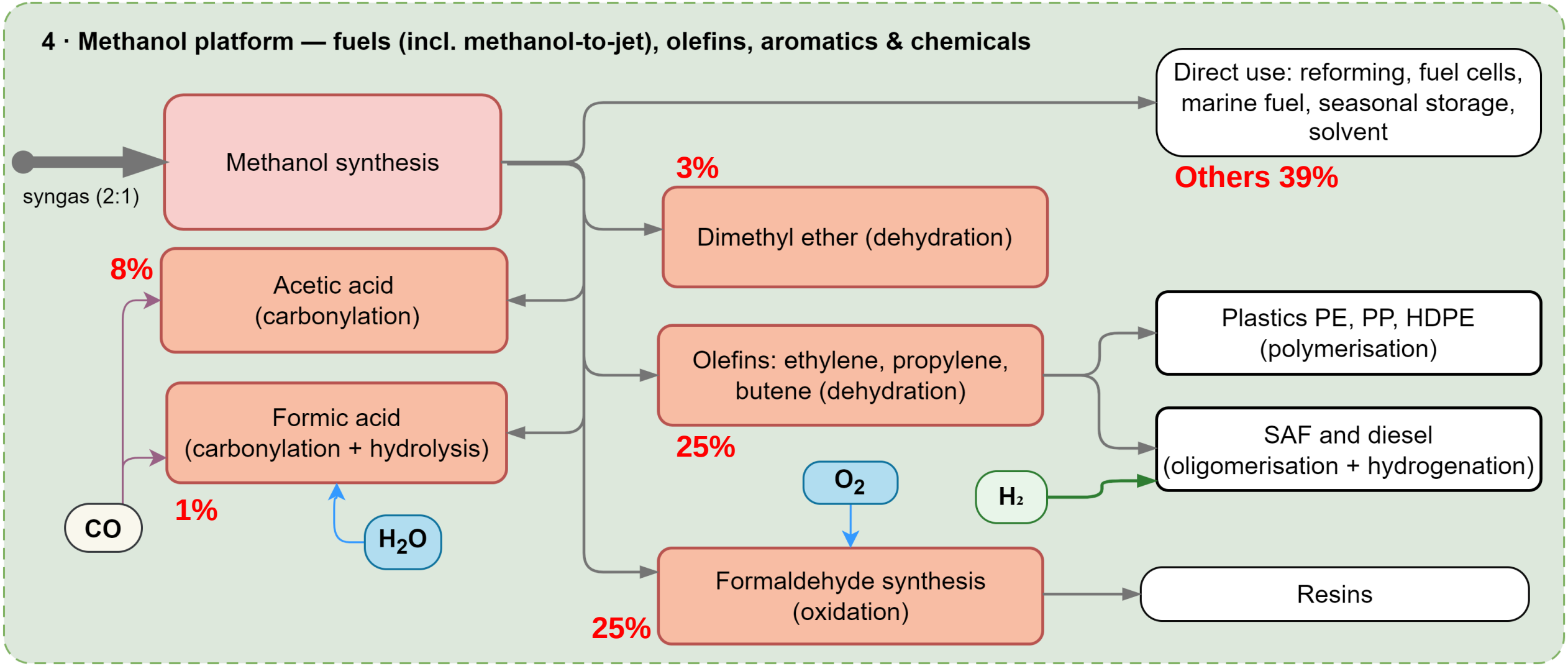
IETS TCP Task XXIV Process Integration for Industry Decarbonization.

IEA / Swiss Federal Office of Energy.

Managers: Daniel Florez-Orrego, François Maréchal.

EPFL team: Oktay Boztas, Meire Ribeiro Domingos, Vibhu Baibhav, Dareen Dardor, Pullah Bhatnagar.

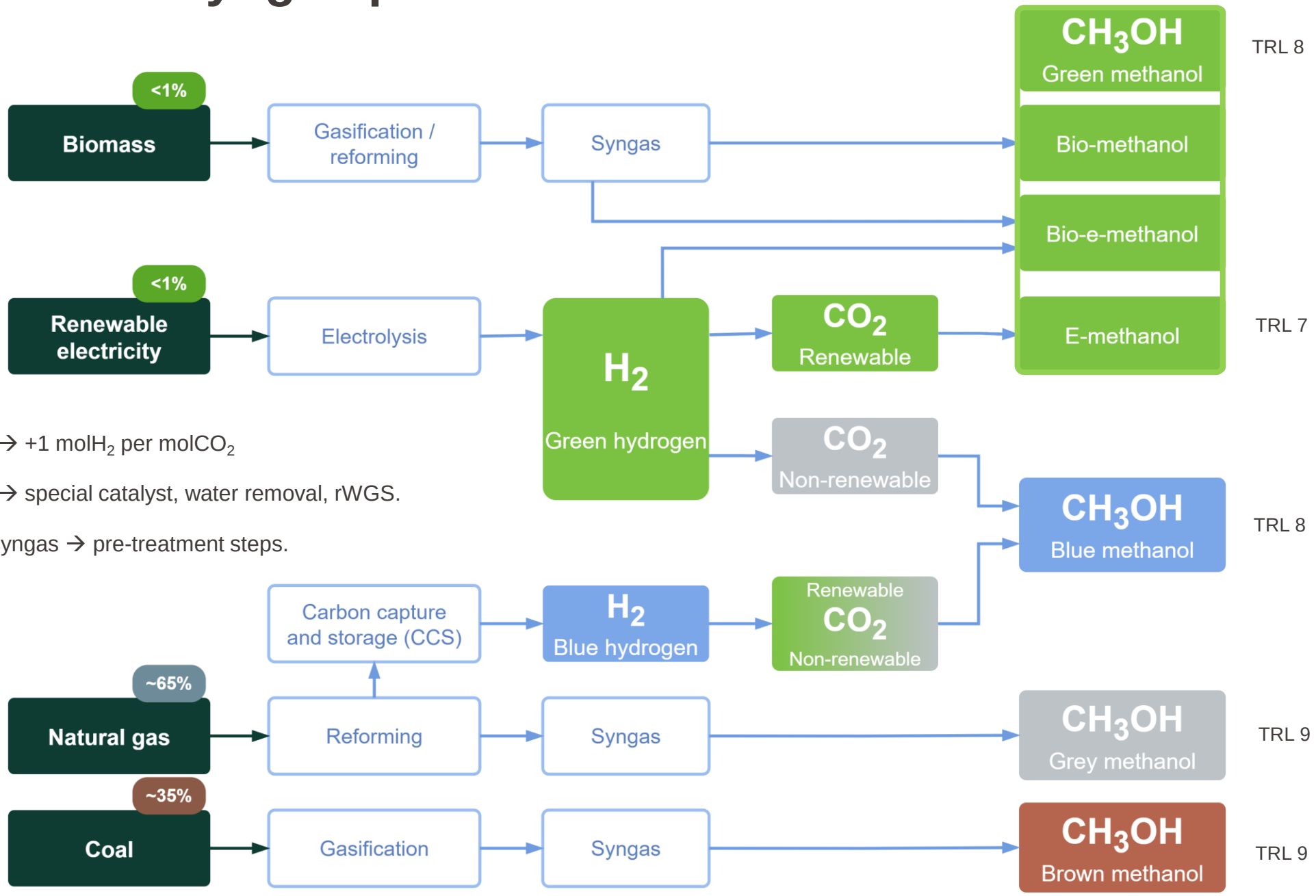
Methanol as platform chemical and energy carrier



Sources for numbers:
Alvarado (2016), IHS Chemical Bulletin, Issue 3
IRENA (2021), Innovation Outlook: Renewable Methanol

More than 60% used in chemicals production

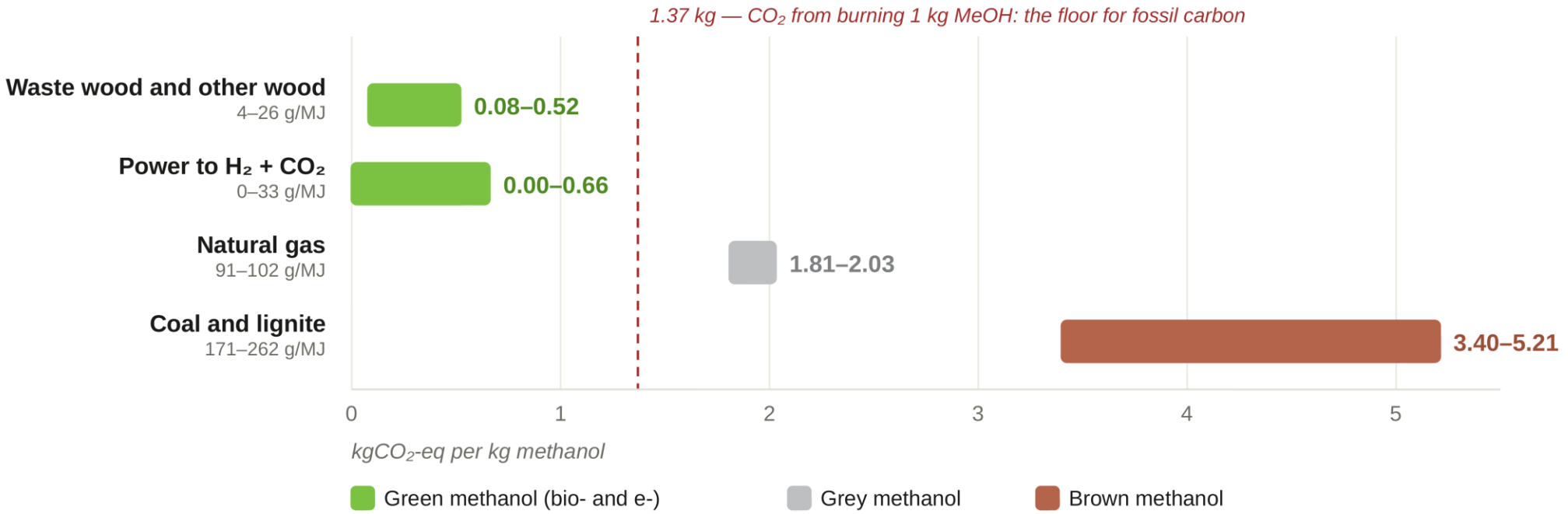
Feedstock for syngas production



- Direct CO_2 hydrogenation → +1 molH_2 per molCO_2
- Direct CO_2 hydrogenation → special catalyst, water removal, rWGS.
- Reforming or gasification syngas → pre-treatment steps.

Sources:
IRENA (2021), Innovation Outlook: Renewable Methanol

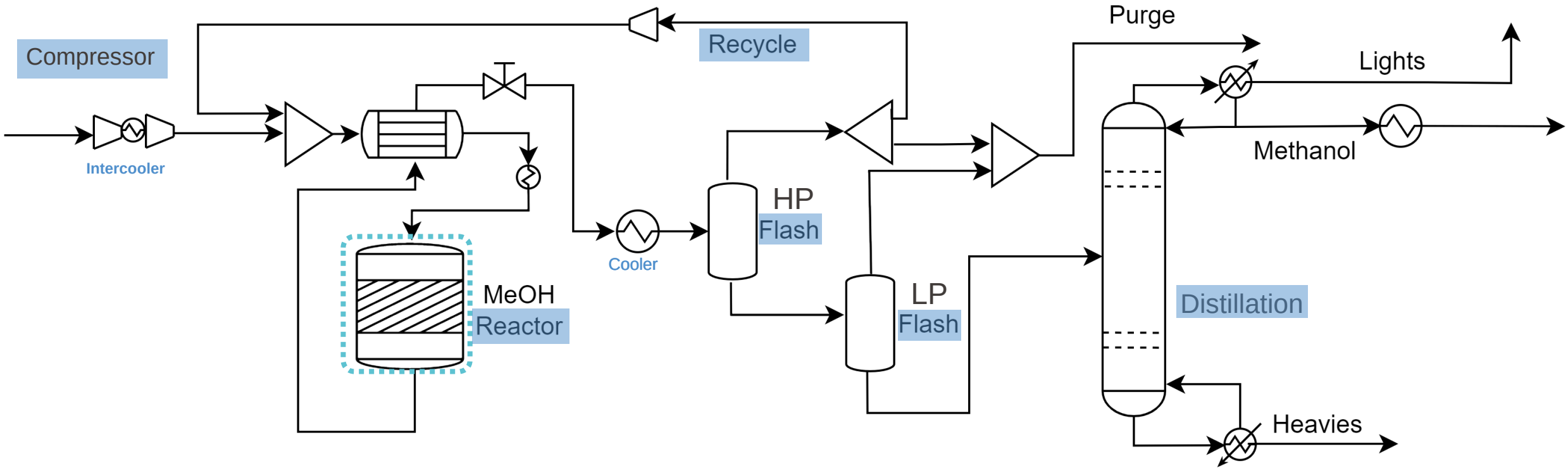
Carbon origin and end of life (EoL)



End-use	Carbon fate	Use	Climate value
Marine fuel, DME, gasoline blend	Combusted → CO_2 back to atmosphere within months	① e-fuel	Avoids fossil extraction; no permanent storage
MTO → ethylene / propylene → PE, PP	Locked in polymers from months to decades. EoL incinerated, recycled or landfilled	② platform chemical	Delayed emissions or permanent storage depending on EoL
MTO → olefins → oligomerisation + hydrogenation → SAF / diesel	Combusted → CO_2 back to atmosphere within months	① e-fuel	Displaces fossil jet fuel; no permanent storage
Formaldehyde, resins, glues, building materials	Locked in construction products for decades	② platform chemical	Decades permanent storage
Acetic acid, MMA, solvents	Stored over months	② platform chemical	Short delay of emissions

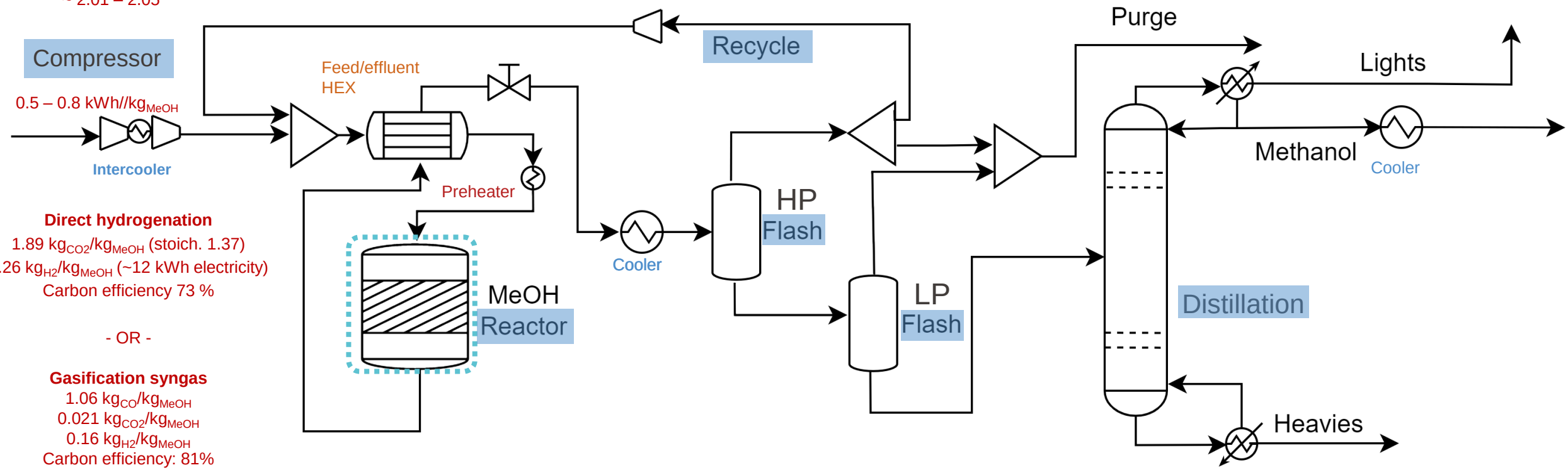
Source: IRENA and Methanol Institute (2021), Innovation Outlook: Renewable Methanol.

Methanol synthesis loop



Methanol synthesis loop

Fresh syngas
stoichiometric number
$(H_2 - CO_2)/(CO + CO_2)$
~ 2.01 – 2.05



Direct hydrogenation
1.89 kg$_{CO2}$/kg$_{MeOH}$ (stoich. 1.37)
0.26 kg$_{H2}$/kg$_{MeOH}$ (~12 kWh electricity)
Carbon efficiency 73 %

- OR -

Gasification syngas
1.06 kg$_{CO}$/kg$_{MeOH}$
0.021 kg$_{CO2}$/kg$_{MeOH}$
0.16 kg$_{H2}$/kg$_{MeOH}$
Carbon efficiency: 81%

Domingos, M.E.G.R., Flórez-Orrego, D., et al. 2022. Comp. & Chem. Eng. 163, 107810.

Methanol synthesis loop

Fresh syngas
stoichiometric number
$(H_2 - CO_2)/(CO + CO_2)$
~ 2.01 – 2.05

Circulator: 0.02 – 0.16 kWh/kg_{MeOH}

Compressor

0.5 – 0.8 kWh//kg_{MeOH}

Intercooler

Feed/effluent
HEX

Recycle

Purge

Lights

Methanol

Cooler

Preheater

HP
Flash

Cooler

LP
Flash

MeOH
Reactor

Distillation

Heavies

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1.89 kg_{CO2}/kg_{MeOH} (stoich. 1.37)
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Carbon efficiency 73 %

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Gasification syngas
1.06 kg_{CO}/kg_{MeOH}
0.021 kg_{CO2}/kg_{MeOH}
0.16 kg_{H2}/kg_{MeOH}
Carbon efficiency: 81%

T: 200-250 °C
P:50-100 bar
Cu / ZnO / Al_2O_3 catalyst - OR -
In-oxide ZrO2-hybrid adv. catalyst
Per-pass conversion <50%

Methanol synthesis reactions

$$CO_2 + H_2 \leftrightarrow CO + H_2O \quad \left(\Delta H^o_{298K} = +41.21\,kJ/mol\right)$$

$$CO + 2H_2 \leftrightarrow CH_3OH \quad \left(\Delta H^o_{298K} = -90.77\,kJ/mol\right)$$

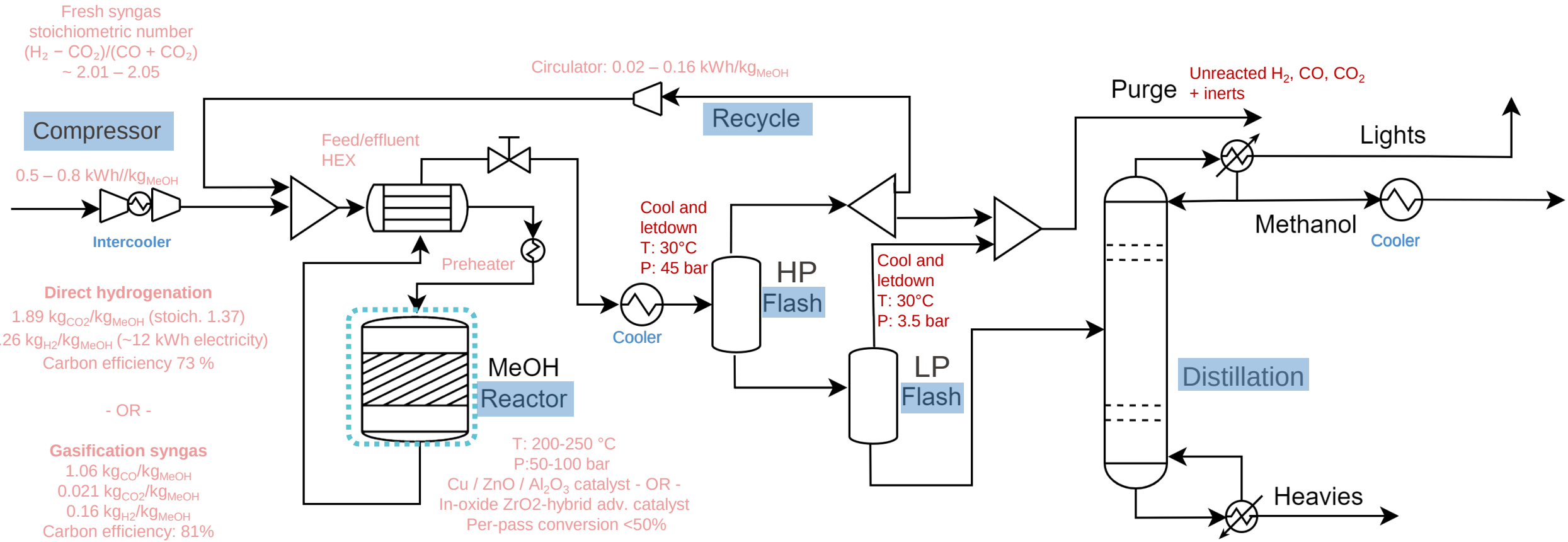
$$CO_2 + 3H_2 \leftrightarrow CH_3OH + H_2O \quad \left(\Delta H^o_{298K} = -49.56\,kJ/mol\right)$$

Waste heat recovery potential

- Reactor heat (~210 °C, steam): 0.4 – 0.8 kWh/kg_{MeOH} → Distillation reboiler ~80 °C: 0.60 – 0.90 kWh/kg_{MeOH}
- Recirculation power: 0.02 – 0.16 kWh/kg_{MeOH} + Fresh syngas compression energy: 0.5 – 0.8 kWh/kg_{MeOH}

Domingos, M.E.G.R., Flórez-Orrego, D., et al. 2022. Comp. & Chem. Eng. 163, 107810.

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MeOH
Reactor

T: 200-250 °C
P:50-100 bar
Cu / ZnO / Al_2O_3 catalyst - OR -
In-oxide ZrO2-hybrid adv. catalyst
Per-pass conversion <50%

Cool and
letdown
T: 30°C
P: 45 bar

Cooler

HP
Flash

Cool and
letdown
T: 30°C
P: 3.5 bar

LP
Flash

Purge

Unreacted H_2, CO, CO_2
+ inerts

CO_2, DME,
methyl formate

Lights

Condenser

Methanol

Cooler

Methanol purity
> 99.85 wt%

Distillation

Heavies

wastewater +
higher alcohols

Reboiler (steam)

Methanol synthesis reactions

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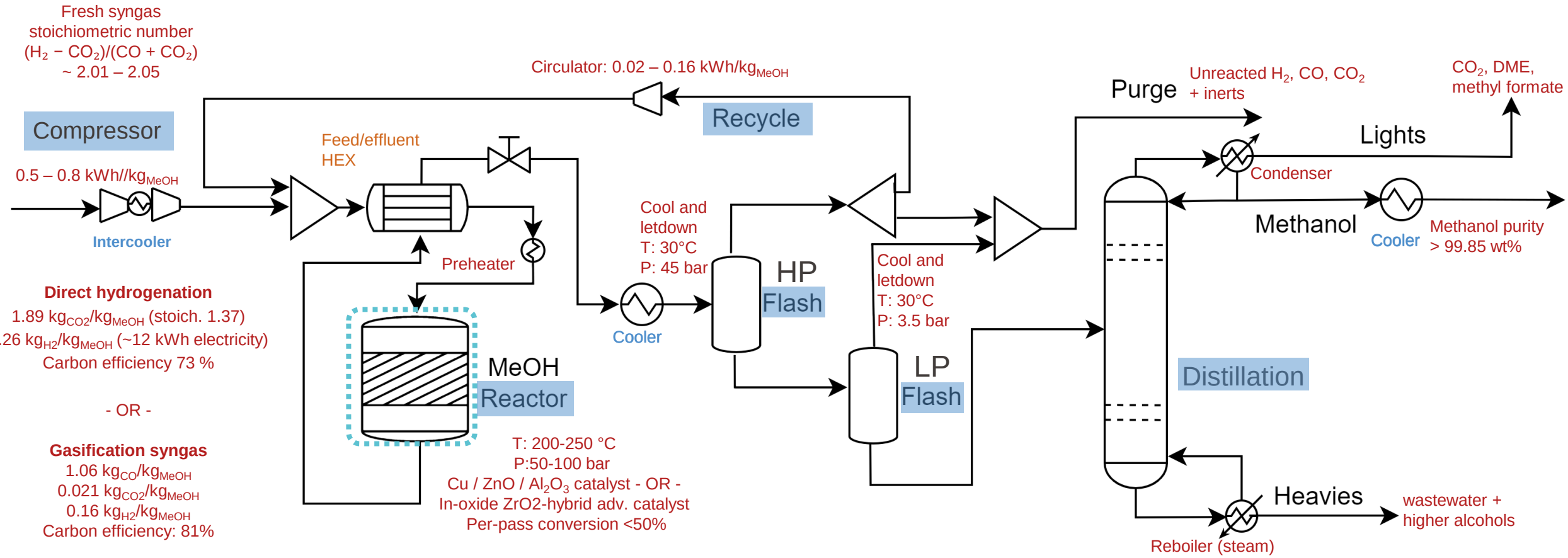
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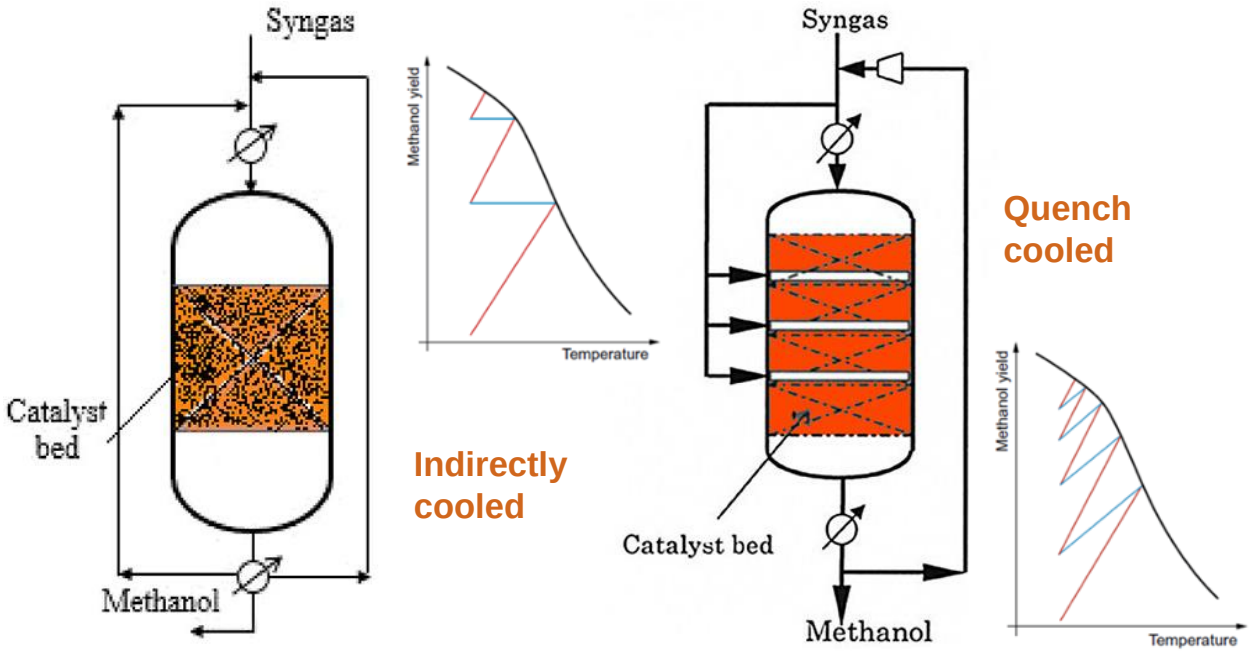
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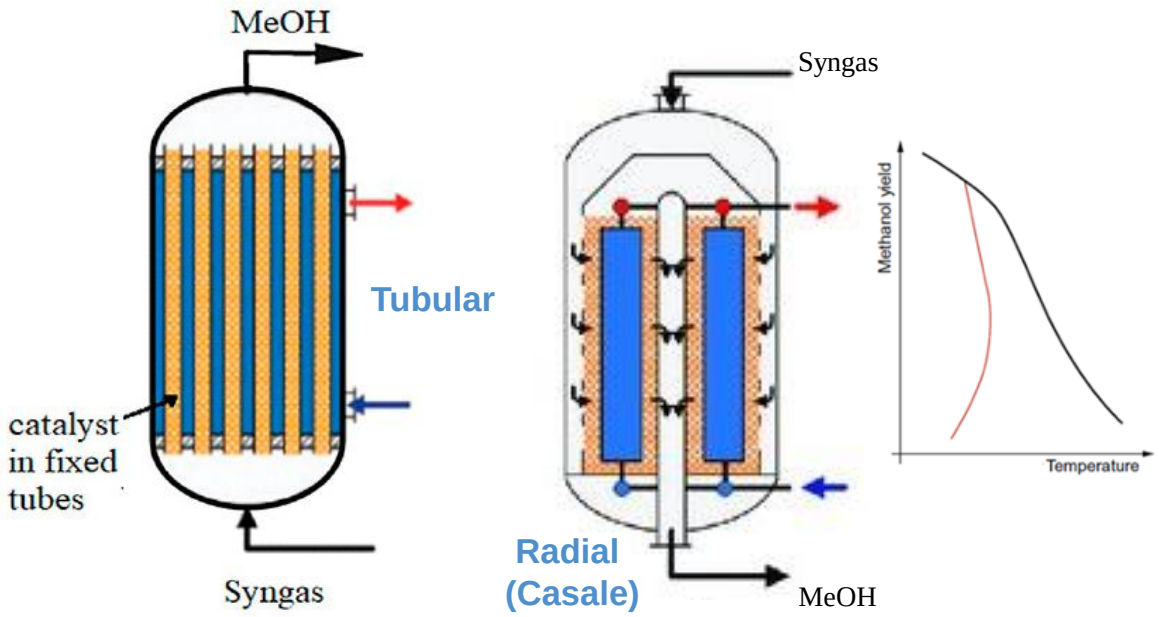
Domingos, M.E.G.R., Flórez-Orrego, D., et al. 2022. Comp. & Chem. Eng. 163, 107810.

Reactor heat management and suppliers

Adiabatic



Isothermal



Parameter	HT ICI	Lurgi	Linde	Toyo	Mitsubishi	AirLiquide Mega Methanol	Casale	AirProducts LPMEOH
Reactor type	Intercooled (adiabatic)	BWR (isothermal)	BWR (isothermal)	BWR (isothermal)	BWR/GCR (isothermal)	BWR/GCR (isothermal)	BWR (isothermal)	Slurry
Pressure (bar)	50–100	50–100	50–150	80–100	55–100	60–75	65–80	30–50
Temperature (°C)	290	270	250	280	270	270	280	215
Conv. per pass (%)	—	36	20-50	60	55–67	>80	—	20–50
Capacity (t/day)	10,000	<2,200	4,000	5,000	—	5,000–10,000	7,000–10,000	<2,000
Industrial plants	>40	>55	8	6	9	>15	9	1 pilot

Dieterich, V. et al, 2020. Power-to-liquid via synthesis of methanol, DME or Fischer–Tropsch-fuels: a review. Energy & Env. Science, 13(10), pp.3207-3252
Bisotti, F. et al. "Impact of Kinetic Models on Methanol Synthesis Reactor Predictions: In Silico Assessment and Comparison with Industrial Data." Ind. Eng. Chem. Res. 2022, 61, 2206–2226

BWR: Boiling Water Reactor / GCRs: Gas-cooled Reactors

e-methanol RFNBO

Kassø

Aabenraa (DK), European Energy / Mitsui & Co.

Status: Operating. First e-methanol plant May 2025
Capacity: 42 kt/y methanol
Electrolysis 52.5 MW, three Siemens Energy units
Biogenic CO_2 Tønder biogas plant; 60 kt/y, liquefaction by Ammongas
Power from adjacent 300 MW Kassø solar park covers 50% demand, 50% from grid
Certification First ISCC EU RFNBO certification for methanol
Offtaker Maersk, LEGO Group, Novo Nordisk





FlagshipONE

Örnsköldsvik (SE), Ørsted, acquired from Liquid Wind

Status Cancelled (2024)
Capacity 55 kt/y methanol (design)
Electrolysis 70 MW
Biogenic CO_2 Hörneborgsverket biomass CHP
Offtaker No long-term offtake at sustainable pricing, significantly higher project costs. The technical case was not the binding constraint.

https://europeanenergy.com/kasso/
https://www.topsoe.com/knowledge-and-insights/flagshipone-client-case
https://www.spglobal.com/energy/en/news-research/latest-news/energy-transition/081524-orsted-scraps-swedish-flagshipone-e-methanol-project-under-development

RFNBO:Renewable Fuels of Non-Biological Origin
ISCC: International Sustainability and Carbon Certification
FID: Final investment decision

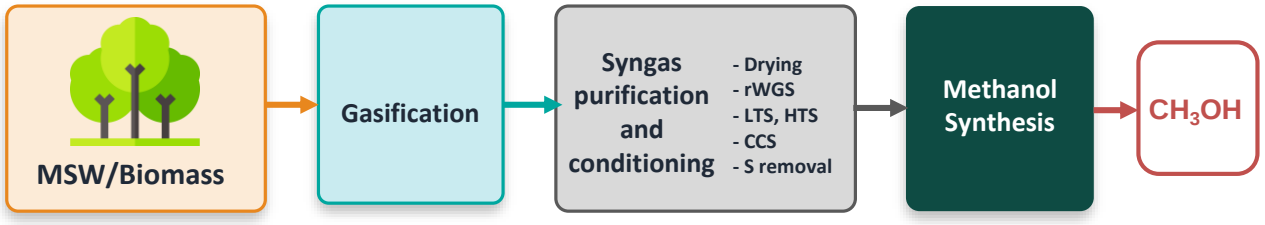
Waste gasification to methanol

Relevant for MSW to syngas facilities

Ecoplanta

El Morell, Tarragona (ES), Repsol + Enerkem + Agbar

Status FID 2025; start-up 2029
Feedstock 400 kt/y non-recyclable MSW reject
Product 240 kt/y methanol
Carbon efficiency ~70 % of feed carbon recovered in the product
GHG ~340 kt/y CO_{2eq} avoided (biogenic)
Investment EUR 800M (incl. EUR 106M EU Innovation Fund grant)
Technology Enerkem gasification; Clariant (guard beds) and MegaMax loop





W2C Rotterdam

Botlek, Port of Rotterdam (NL), Enerkem + Shell + partners

Status Announced 2019; repurposed to Waste-to-Jet in 2021.
Feedstock 360 kt/y non-recyclable mixed waste
Product 220 kt/y bio-methanol (original scope)
GHG ~ 300 kt/y CO_2 avoided (biogenic)
Configuration 2 Enerkem lines; O_2 from Air Liquide, H_2 from Nouryon
Relevance Abandoned as a methanol project before FID

https://www.repsol.com/en/sustainability/sustainability-pillars/environment/circular-economy/our-projects/transforming-waste-into-chemicals/index.cshtml
https://www.portofrotterdam.com/en/news-and-press-releases/w2c-rotterdam-project-welcomes-shell-partner

e-methanol from green H_2 and CO_2 at NZV-scale

Stream	Input/Output	Specific value	Average flow	Annual flow (8760 h/y)
MATERIAL		**kg per kg_{MeOH}**	**t/h**	**kt/y**
Carbon dioxide	feedstock	1.89	37.8	**330**
Hydrogen	feedstock	0.25	5.0	44
Methanol	product	1.00	20.0	**175**
Wastewater + Purge	by-product	0.74 + 0.39	14.9 + 7.9	130 + 69
ENERGY		**kWh per kg_{MeOH}**	**MW**	**GWh/y**
Electricity, electrolyzer	power	12.0	240	**2100**
Electricity, compression	power	1.0	20	180
Recoverable heat (130 → 30 °C)	waste heat	2.0	40	351

NZV 480-960 t_{MeOH}/d → 20-40% of a single 2200 t_{MeOH}/d Lurgi methanol plant & 12-22x electrolyzer modules of 20 MW

Stoich 1.375 kg_{CO2} per kg_{MeOH}

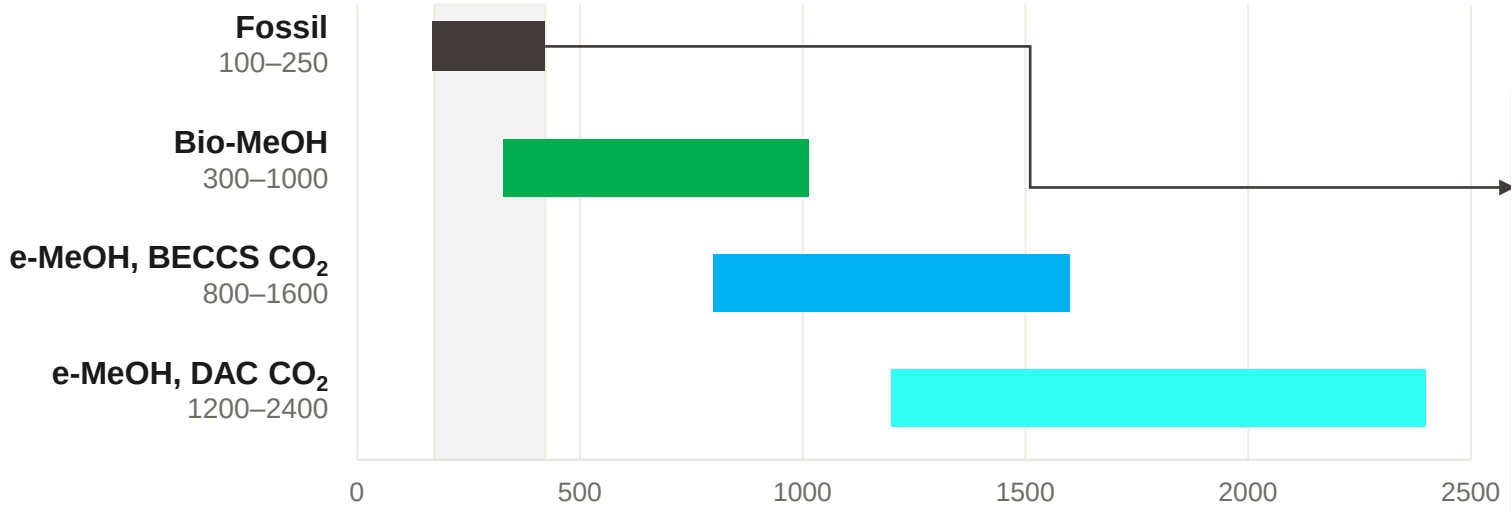
Florez-Orrego, Ribeiro Domingos, M., D., Boztas, D., Baibhav, V., et al 2026, Renewable energy integration and waste heat valorization in heavy industry for the co-production of fuels and chemicals. DECHEMA Forum 2026, Frankfurt, Germany

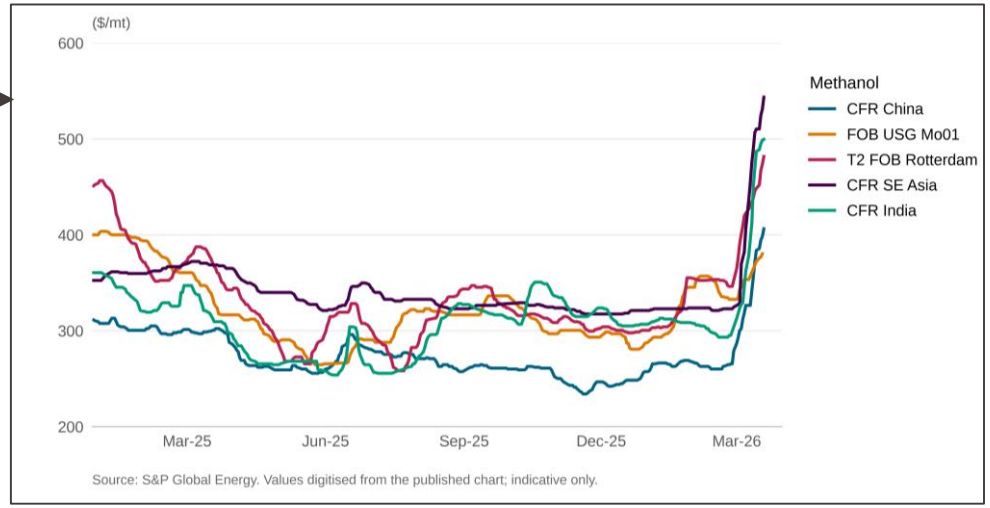
Methanol production costs and levers

2 - 4×

vs fossil market price of 200–400 EUR/t_{MeOH}

Production cost by route (EUR/t_{MeOH})





What drives the cost

- **H_2 and power 55–65%**: 55 kWh/kg_{H2}
- **CAPEX 15–25%**: may falls 5× from demo to Mt/y scale.
- **CO_2 feedstock 5–10%**: 30–60 EUR/t from point sources vs >600 EUR/t from DAC
- **OPEX and O&M 5–10%**: partly offset by O_2 selling and low-grade heat use in DHN
- **Auxiliary power 3–4%**

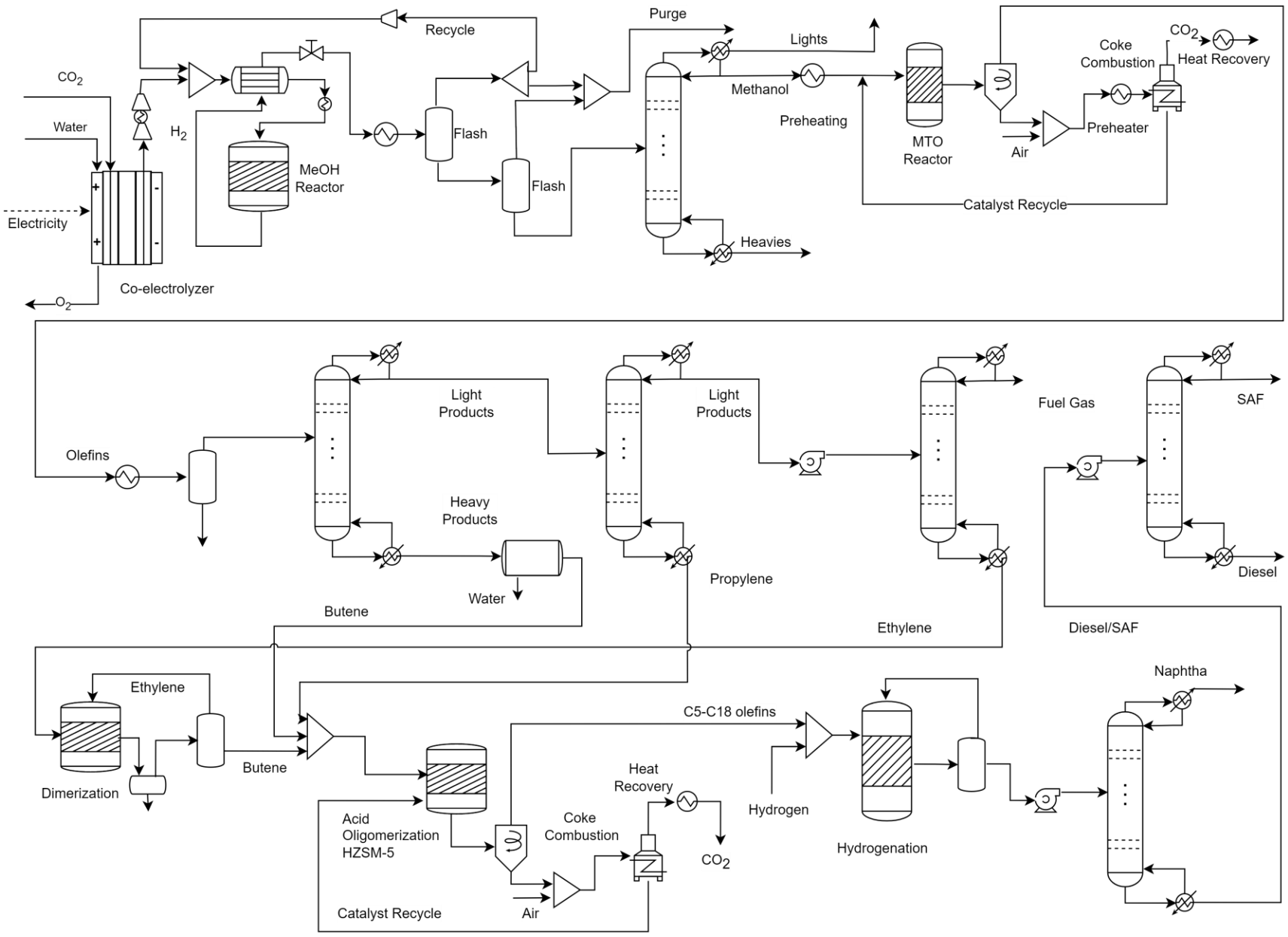
What can help close the gap

- Multi-year offtake at >1000 EURO/$t_{MeOH-eq}$ indexed on FuelEU compliance.
- Electricity purchase via long-term PPA at low value (<60 EUR/kWh)
- RefuelEU Aviation/RED III RFNBO: binding renewable-fuel quotas
- CBAM from 2026 imported fossil priced chem. at EU CO_2 value (MeOH not yet)
- Contracts for difference and H_2Global-type auctions (subsidies H_2)

Source: IRENA and Methanol Institute (2021), Innovation Outlook: Renewable Methanol.

CBAM: Carbon Border Adjustment Mechanism

CO_2 utilization route to jet blendstock



	e-Jet blendstock
Electricity	**700 GWh/y (80 MW)**
H_2O	23 t/h
Jet blendstock	2.73 t/h
Diesel	0.26 t/h
Naphtha	0.70 t/h
CO_2	**122 kt/y (15.3 t/h)**
O_2 (surplus)	12.2 t/h
Waste heat	28 MW

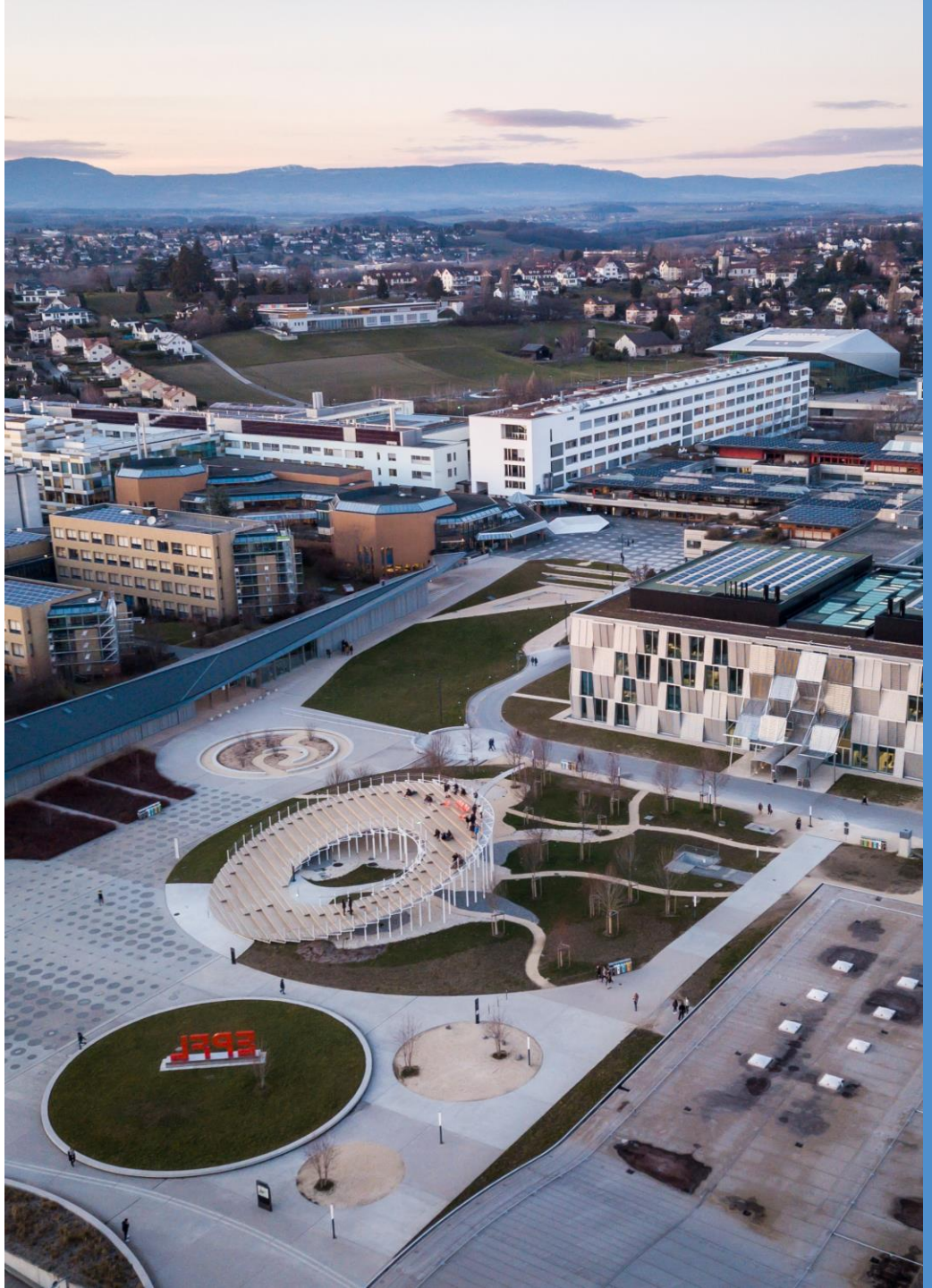
Jet selectivity (>70%) of liquid product slate, higher than Fischer Tropsch (35%), at the expense of 30 kWh/kg of jet

European reference prices are 2.09 EUR/kg for bio-SAF and 7.70 EUR/kg for synthetic e-SAF, against 0.73 EUR/kg for conventional jet fuel.

Florez-Orrego, Ribeiro Domingos, M., D., Boztas, D., Baibhav, V., et al 2026, Renewable energy integration and waste heat valorization in heavy industry for the co-production of fuels and chemicals. DECHEMA Forum 2026, Frankfurt, Germany

Some challenges for an NZV e-methanol projects

Continuous CO_2 supply	Point sources at 10–30 vol% CO_2 upgraded to 95 vol% CO_2 after capture at < 80 EUR/t. Waste-derived CO2 is only partly biogenic. Buffer CO2 storage to allow for industrial emissions downtime.
Low-carbon power for green H_2 supply	Hydropower base-load with larger PV delivers the lowest carbon intensity profile. At that scale, it is a grid-connection challenge. Electrification of heat competes with chemical plants. At 0.24 kg H2 per kg MeOH, electrolyser efficiency and power price set most of the product cost.
Seasonal mismatch in power and heat demands	Summer brings cheap power, but small heat demand in DHN; in winter it is the opposite.
Feedstock impurities	Waste-derived gas carries halogens, sulphur and metals that poison catalysts. Stringent syngas pre-treatment becomes mandatory.
Cost premium and offtaker	Cost is 2–4x fossil: Lonza (acetic acid, formaldehyde, solvents)? Metafuels (methanol-to-jet)? Syngenta (industrial solvents)? or external port offtake for shipping fuels? Certified demand may outpace supply and hold prices up.
Timing and uncertainties	Long project horizon: ~4 years from FID to start-up (Ecoplanta), with e-methanol still at TRL 7. RFNBO rules (RED II/III, IMO) are still evolving. No premium is contracted yet.
Exploratory approaches	European Union projects to exchange, learn and mitigate the risk? Seek for small demonstrator or target a joint larger plant inside NZV.



Industrial cases